

The MoCHII Atomic-Data Fitting Pipeline:

CHIANTI Cooling and n -level Fits

Abstract

MoCHII evaluates all metal-line atomic data from analytic fits; the raw CHIANTI database is never parsed at runtime. The Python pipeline under `tools/fitting/` is the auditable source of every coefficient in `data/atomic/`. This memo records the first production cycle (2026-07-11) for the ion set O II, O III, N II, S II, S III, Ne II, and Ne III: the cooling-function fits used inside the thermal iteration, the n -level data files (level energies, A -values, and Chebyshev fits of the effective collision strengths in Burgess–Tully scaled space) used for output-time line emissivities, and the verification of both against the CHIANTI v11.0.2 source data and against PyNeb. The fitted files reproduce direct CHIANTI evaluation to $\lesssim 0.5\%$ in the classic diagnostic ratios; the remaining differences from PyNeb (up to 6–10% for the density diagnostics near their critical densities) trace to PyNeb’s independent choice of collision strengths and A -values, not to fit error.

1. Overview: two products, one design rule

The metal-line design requires that adding an ion be a data operation, not a code operation. The pipeline therefore produces, for each registered ion, self-contained coefficient files with provenance headers (CHIANTI version, population model, fit range, fit form, maximum fit error), of two kinds:

- **Cooling fits (hot loop).** A smooth closed-form cooling function $\Lambda_{\text{ion}}(T)$, cheap and Newton-friendly inside the thermal-balance iteration. Coronal (low-density) limit by construction.
- **n -level data (diagnostics).** Level energies, statistical weights, radiative A -values, and fitted effective collision strengths $\Upsilon(T)$ feeding the generic n -level solver, evaluated only at output time on the converged state. These supply the density-dependent emissivities and the diagnostic line ratios that the cooling fits cannot give.

The pipeline is seeded from EXHALE: `chianti_cooling.py` (copied from EXHALE/cooling_data/) reads the raw CHIANTI ASCII ion files (`.elvlc`, `.wgfa`, `.scups`) directly, with no ChiantiPy dependency, and applies the Burgess–Tully (1992) descaling of the stored scaled collision strengths. CHIANTI v11.0.2 lives at `~/.RT_Codes/CHIANTI/dbase`.

2. The cooling function

For a transition $l \rightarrow u$ with energy gap ΔE_{lu} , the Maxwellian electron-impact excitation rate coefficient is

$$q_{lu}(T) = \frac{8.629 \times 10^{-6}}{g_l \sqrt{T}} \Upsilon_{lu}(T) e^{-\Delta E_{lu}/kT} \quad [\text{cm}^3 \text{s}^{-1}], \quad (1)$$

with $\Upsilon_{lu}(T)$ the effective collision strength obtained by descaling the `.scups` data. In the optically thin, low-density limit every collisional excitation is followed by a radiative decay, so the line cooling normalized by $n_e n_{\text{ion}}$ is

$$\Lambda_{\text{ion}}(T) = \sum_{l \rightarrow u} f_l(T) q_{lu}(T) \Delta E_{lu} \quad [\text{erg cm}^3 \text{s}^{-1}], \quad (2)$$

where $f_i(T)$ is the fractional population of the lower level. The pipeline uses the *ground-term* population model: the levels within 0.4 eV of ground (the ground-term fine structure, e.g. the $^3P_{0,1,2}$ block of O III) are Boltzmann-distributed and all excitations out of them cool. For the single $^4S_{3/2}$ ground level of O II and S II this reduces to the coronal limit.

3. Cooling-function fits

Each $\Lambda_{\text{ion}}(T)$ curve is fitted over 10^3 – 10^5 K with the multi-exponential form used throughout EXHALE,

$$\Lambda_{\text{ion}}(T) = T^{-1/2} \sum_{i=1}^N A_i e^{-T_i/T}, \quad (3)$$

a Levenberg–Marquardt fit on $\log \Lambda$ with the T_i initialized from the level-energy ladder of each ion (fine structure / forbidden optical / UV blocks read off the `.elvlc` file). The fitter tries $N, N+1, N+2$ terms and keeps the best. One case needed care: Ne II cooling is a single fine-structure line ($12.8 \mu\text{m}$, $\Delta E/k \simeq 1123$ K) whose slow decay ($\Lambda \propto T^{-0.35}$, set by the T dependence of Υ) must be superposed from a ladder of T_i values before the far UV block at $\sim 3 \times 10^5$ K takes over; it converges with $N = 7$. Table 1 lists the fit quality; Figure 1 overlays the fits on the CHIANTI curves. The coefficients are written to `data/atomic/cooling_<ion>.txt`.

Table 1: Cooling fits, 10^3 – 10^5 K. The maximum relative error is against the direct CHIANTI evaluation of Eq. (2).

Ion	N	max error	output file
O II	5	0.07%	cooling_o_2.txt
O III	4	1.83%	cooling_o_3.txt
N II	6	0.14%	cooling_n_2.txt
S II	3	2.26%	cooling_s_2.txt
S III	6	0.46%	cooling_s_3.txt
Ne II	7	0.65%	cooling_ne_2.txt
Ne III	4	0.58%	cooling_ne_3.txt

A caveat carried in every file header: Eq. (3) is low-density-limit cooling. Transitions with low critical density (the fine-structure IR lines; [O II] $\lambda\lambda 3726, 3729$; [S II] $\lambda\lambda 6717, 6731$) saturate above n_{crit} , where the cooling fit overestimates the cooling. If an application demands it, a fitted density-suppression factor can be added later; the emissivities themselves always come from the n -level data.

4. n -level data with fitted collision strengths

For each ion the lowest n levels are exported — enough for the optical/IR diagnostics (Table 2); extra levels are harmless to the solver. The file `data/atomic/nlevel_<ion>.txt` contains three blocks:

- NLEV: level index, statistical weight g , energy in cm^{-1} (observed values preferred, theoretical as fallback);
- NRAD: radiative rates A_{ul} from `.wgfa` (duplicate rows summed);

- NUPS: one line for each collisional transition — the Burgess–Tully type, scaling parameter C , transition energy in Ryd, the fitted domain $[s_{\text{lo}}, s_{\text{hi}}]$, a log flag, and the Chebyshev coefficients, followed by the recorded maximum fit error.

4.1 Burgess–Tully scaling and Chebyshev fits

CHIANTI stores Υ on the Burgess–Tully (1992) scaled temperature $s \in [0, 1]$. With $e_t \equiv kT/\Delta E$, the forward map is

$$s = \begin{cases} 1 - \frac{\ln C}{\ln(e_t + C)}, & \text{types 1, 4,} \\ \frac{e_t}{e_t + C}, & \text{types 2, 3, 5, 6,} \end{cases} \quad (4)$$

and the descaling back to Υ is

$$\Upsilon = y \ln(e_t + e) [1]; \quad y [2]; \quad \frac{y}{e_t + 1} [3]; \quad y \ln(e_t + C) [4]; \quad \frac{y}{e_t} [5]; \quad 10^y [6], \quad (5)$$

where $y(s)$ is the scaled value. Instead of the spline-through-table evaluation used by CHIANTI itself, the pipeline fits $y(s)$ with a Chebyshev series over the image of $T \in [10^3, 10^5]$ K intersected with the tabulated range,

$$u = \frac{2(s - s_{\text{lo}})}{s_{\text{hi}} - s_{\text{lo}}} - 1, \quad y = \sum_{k=0}^{n_c-1} c_k T_k(u), \quad (6)$$

clipping s into $[s_{\text{lo}}, s_{\text{hi}}]$ outside the fit domain. The degree grows from 2 until the descaled $\Upsilon(T)$ matches the direct spline evaluation to 0.2% (cap at 12). Weak intercombination transitions whose tables span decades (e.g. O III $^1D_2, ^1S_0 \rightarrow ^5S_2$) defeat a linear-space fit; for those the series stores $\log_{10} y$ instead, marked by the log flag (logf=1, apply 10^y before descaling). The Fortran `nlevel_mod` must evaluate exactly this recipe; it is replicated in `verify_nlevel_pyneb.py` as the reference implementation.

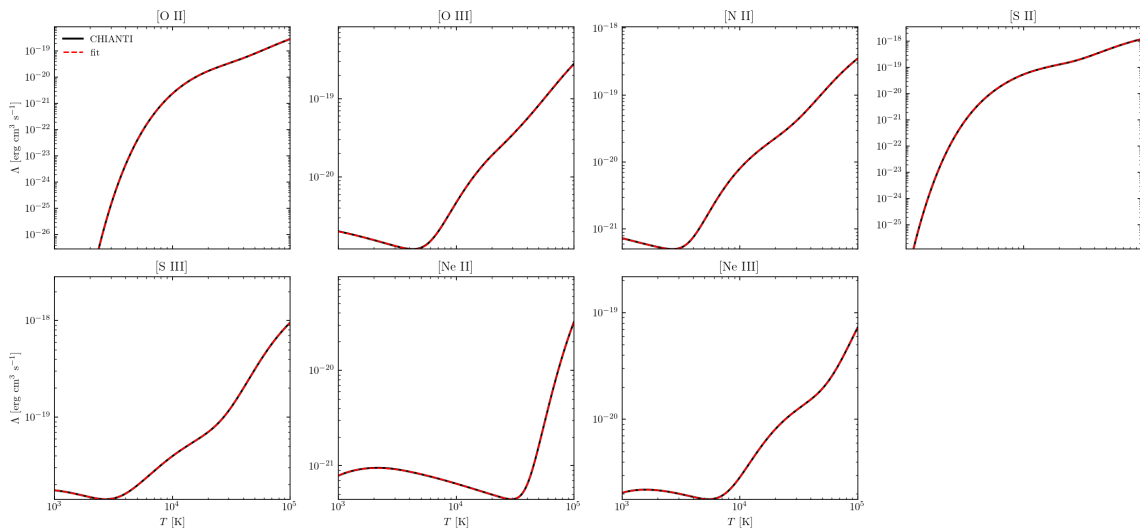


Figure 1: Cooling fits (red dashed) over the direct CHIANTI curves (black), 10^3 – 10^5 K, ground-term population, low-density limit.

Table 2: n -level exports. “worst Υ error” is the largest maximum relative error among the fitted transitions of that ion, measured on the descaled $\Upsilon(T)$ against the direct spline evaluation over 10^3 – 10^5 K.

Ion	levels	A -values	Υ fits	worst Υ error
O II	5	9	10	0.28%
O III	6	10	15	1.68%
N II	5	8	10	0.17%
S II	5	9	10	1.25%
S III	6	12	15	0.29%
Ne II	2	1	1	0.16%
Ne III	5	9	10	0.19%

The two entries above 1% have benign origins: the O III outlier is the physically negligible 1S_0 – 5S_2 intercombination transition ($\Upsilon \sim 10^{-4}$), and the S II errors reflect the coarse 10-point .scups tables of that ion — the deviation is from the interpolating spline, at the level of the resolution of the source data itself.

4.2 Statistical equilibrium and emissivities

Given (T_e, n_e) , the solver builds the rate matrix from the file contents alone: collisional rates from Eq. (1) with de-excitation by detailed balance, $q_{ul} = 8.629 \times 10^{-6} \Upsilon / (g_u \sqrt{T})$ and $q_{lu} = (g_u / g_l) q_{ul} e^{-\Delta E / kT}$, plus the radiative A_{ul} ; the level populations follow from $\dot{n} = 0$ with the closure $\sum_i n_i = 1$, and the emissivity of a line is $j_{ul} \propto n_u A_{ul} \Delta E_{ul}$.

5. Verification

5.1 Internal consistency: fitted files vs. direct CHIANTI

Running the same statistical-equilibrium solver twice — once reading the fitted `nlevel` files, once evaluating the raw CHIANTI splines and .wgfa rates directly — isolates the error contributed by the fits and the file format. Over the grid $T_e \in \{5, 10, 20\} \times 10^3$ K, $n_e \in \{10^2, 10^3, 10^4\} \text{ cm}^{-3}$:

Diagnostic ratio	worst fit/raw − 1
[O II] $\lambda 3726 / \lambda 3729$	0.05%
[S II] $\lambda 6717 / \lambda 6731$	0.52%

The fitted files are faithful to CHIANTI v11.0.2.

5.2 Cross-check against PyNeb

PyNeb (v1.1.31) carries its own atomic data — for these ions the collision strengths SSB14 (O III), T11 (N II), Kal09 (O II), TZ10 (S II) with A -values FFT04-SZ00, FFT04, Z82-WFD96, RGJ19 respectively — so this comparison is a cross-calibration against an independent reference, not a fit-residual test. Tables 3 and 4 give the classic temperature and density diagnostics.

Table 3: Temperature diagnostics vs. PyNeb.

T_e [K]	n_e [cm ⁻³]	[O III] $\lambda 5007/\lambda 4363$			[N II] $\lambda 6584/\lambda 5755$		
		MoCHII	PyNeb	diff	MoCHII	PyNeb	diff
5000	10 ²	4104	4153	-1.2%	829.6	828.6	+0.1%
5000	10 ³	4093	4133	-1.0%	818.3	817.2	+0.1%
5000	10 ⁴	3907	3943	-0.9%	624.7	623.9	+0.1%
10000	10 ²	152.0	153.4	-0.9%	66.97	66.93	+0.1%
10000	10 ³	151.7	152.8	-0.7%	65.75	65.64	+0.2%
10000	10 ⁴	146.0	147.0	-0.6%	52.23	52.12	+0.2%
15000	10 ²	50.97	51.30	-0.6%	29.06	29.08	-0.1%
15000	10 ³	50.77	51.10	-0.6%	28.53	28.59	-0.2%
15000	10 ⁴	49.08	49.43	-0.7%	23.26	23.32	-0.3%
20000	10 ²	29.82	29.78	+0.2%	19.23	19.23	+0.0%
20000	10 ³	29.58	29.67	-0.3%	18.92	18.92	-0.0%
20000	10 ⁴	28.68	28.81	-0.5%	15.71	15.71	+0.0%

Table 4: Density diagnostics vs. PyNeb.

T_e [K]	n_e [cm ⁻³]	[O II] $\lambda 3726/\lambda 3729$			[S II] $\lambda 6717/\lambda 6731$		
		MoCHII	PyNeb	diff	MoCHII	PyNeb	diff
5000	10 ²	0.8045	0.7641	+5.3%	1.298	1.333	-2.6%
5000	10 ³	1.436	1.383	+3.9%	0.8223	0.8396	-2.1%
5000	10 ⁴	2.364	2.491	-5.1%	0.5577	0.5052	+10.4%
10000	10 ²	0.7697	0.7411	+3.9%	1.309	1.351	-3.1%
10000	10 ³	1.260	1.241	+1.5%	0.8810	0.9139	-3.6%
10000	10 ⁴	2.267	2.397	-5.4%	0.5655	0.5203	+8.7%
15000	10 ²	0.7566	0.7341	+3.1%	1.320	1.349	-2.1%
15000	10 ³	1.180	1.179	+0.2%	0.9185	0.9520	-3.5%
15000	10 ⁴	2.209	2.345	-5.8%	0.5704	0.5294	+7.7%
20000	10 ²	0.7502	0.7319	+2.5%	1.343	1.347	-0.3%
20000	10 ³	1.136	1.144	-0.7%	0.9543	0.9786	-2.5%
20000	10 ⁴	2.170	2.311	-6.1%	0.5766	0.5373	+7.3%

The temperature diagnostics agree to 1.2% ([O III]) and 0.3% ([N II]). The density diagnostics drift to 6–10% at $n_e = 10^4 \text{ cm}^{-3}$, near and above the critical densities of the 2D levels, where the ratios become A -value-limited — and the two codes take their A -values from different compilations (CHIANTI v11 vs. Z82-WFD96 and RGJ19). Since the fitted files reproduce raw CHIANTI to $\leq 0.5\%$ (previous subsection), this spread measures the state of the atomic data, not the pipeline; it should be kept in mind as the floor of attainable agreement when the validation gates compare MoCHII line ratios with PyNeb-based references.

6. Adding an ion

The procedure is a data operation:

1. add one entry (element, stage, level-energy guesses / level count) to the IONS dictionary of `fit_cooling.py` and of `fit_nlevel.py`;
2. run both scripts; check the reported maximum fit errors and the overlay figure;
3. extend `verify_nlevel_pyneb.py` with the diagnostic ratios of the new ion where PyNeb has data, and record the comparison.

The suggested order of addition (cooling importance plus diagnostics) continues with C, Ar, and Fe.